Operator tool for the CIR website: requirements

Status: draft for the CIR team, reviewed 17 October 2026. **Basis:** the CIR website template at commit `23f4878b60d337ebd18fac22427dcd31bdf14a86`, read file by file (nothing built or run), and the existing operator tool in this repository (`site-agent` v0.1, events only). Facts about the template are in EDITABLE_SECTIONS.md. Line numbers there and here refer to the files named.

At a glance

- The operator changes content through buttons, dropdowns and short forms. A local model drafts wording from facts the operator gives it. Nothing is saved until the operator approves a preview.
- Only content changes. Pages, colours, navigation shape, research directions and the list of content types stay as they are. Structure changes need a developer.
- The template cannot yet be trusted with all content. Three problems must be fixed first (proposed changes P1, P2 and P3 in section (e)): most page wording is written into code files; partners, funding and research cannot be marked as samples, so placeholders there are invisible to the launch check; and a few values silently break links or lists.
- The existing operator tool is a good base. Its default permission to edit code files (`*.astro`, `*.ts`) must be removed, because it would let the tool change the site's structure.
- Windows and WSL2 both need a test on the actual machine before a bundle is committed to. The template's installed packages differ by operating system (`package-lock.json`).

Words used in this document. *Record*: one content file with fields, for example one person. *Build*: turning the site files into the public website. *Check*: the template's own check command (build plus HTML check). *Launch check*: the template's readiness list. *Model*: the local language model that drafts wording. *Ollama*: the program that runs the model on the computer. *WSL2*: Windows Subsystem for Linux, version 2. *Approve*: the operator sees the exact changes and clicks Approve. *Block*: the tool refuses to save. *Warn*: the tool shows a warning and the operator may continue.

(a) Operator personas and knowledge level

Persona	Who	What they know	What they do with the tool	What they must not do
Operator (main user)	Lab or centre staff member	The lab, its people, events and processes. No HTML, Astro, npm or git. Comfortable with Windows, email and word processing. May never have used a terminal	Adds and edits records; reviews drafts; approves changes; runs the check; undoes a change; checks the launch list	Edit code or settings files; publish to the live site; set the domain; approve legal or finance text
Content approver	Head of centre or a faculty lead	What the centre is willing to say in public. Can read a preview	Approves wording, research descriptions and photographs (with consent). Signs off the launch list	Needs no technical skill; must not change structure
Institute reviewers	Legal office, finance office, data-protection contact	Their own policies	Read the legal, privacy, tax and donor text in preview. Give approval in writing; the approval is recorded in the tool	Edit files. Draft text is never produced by the model for them

Persona	Who	What they know	What they do with the tool	What they must not do
Technical contact	Developer or IT support	Node, npm, WSL2, Windows administration, Ollama, git	Installs the tool and the model; sets the domain and the club link; deploys the built site; upgrades the template; removes the legal “Draft for review” notice after approval	Change the template’s structure without a review
Website visitor	The public	Varies: phones, screen readers, slow connections	Does not use the tool. Their experience is the test: phone width, keyboard use and clear wording	Not applicable

(b) Use cases

Each row is one thing the operator might need to do. “Model drafts” lists only text the model may write, from facts the operator supplies. Every use case that writes content ends with an approval step.

B1. People

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U1	Add a person. A new member joins, or a former member is to be listed	Name; group (dropdown of six); role; department; interests; homepage (optional); portrait (optional); three to six facts for the profile; consent tick if a portrait is used	Biography, 80 to 150 words, from the facts only	Name present and not already listed (capitals and spacing ignored). Group from the list. Homepage is https. Portrait is JPG or PNG, at least 800 px tall. Biography contains no name, year, number or link that is not in the facts	Card and profile preview; the biography; the consent tick	New person record; portrait copied to the portraits folder under the person’s short name
U2	Edit a person. Role, department or interests change	Choose the person; change fields	Optional new biography from new facts	As U1. Project links re-checked	Changed lines shown before and after	Only the changed lines are rewritten
U3	Move a person to an alumni group. The person has left. (This is the template’s only “archive”)	Choose the person; choose the alumni group	None	Warns that the person stays named in their projects	Confirm the move	Group changed; person moves to the alumni list
U4	Remove a person. Entered by mistake, or a sample is removed	Choose the person	None	Lists projects that name them (those names become plain text). Requires a typed confirmation	Confirm	Record deleted; a copy kept in the run backup

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U5	Add or replace a portrait	Image file; consent tick	None	JPG or PNG; at least 800 px tall; warns if not close to 4:5; size limit; file name from the person's short name	Crop preview	Portrait saved; the person's photo field set

B2. Student projects and research directions

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U6	Add a student project. A project is approved	Title; direction (five choices); students (pick from People, or type a name); status (optional); summary facts; featured yes or no	Summary, up to 30 words, from the title and facts	Title up to 120 characters. Each student matched to a person, else a warning with the closest names. At most four featured (warning). Direction from the list	Card and project page preview	Project record created
U7	Edit a project. Students, status or direction change	Choose the project; change fields	None	As U6	Changed lines shown	Only changed lines rewritten
U8	Write or replace an abstract	Notes, or a draft abstract	Abstract, 120 to 250 words, from the notes	Word count. No new names, numbers or dates. Placeholder text removed	Approve the text	Abstract saved
U9	Feature or unfeature a project for the homepage	Yes or no; order	None	Warns when more than four are featured	Approve	Featured setting saved
U10	Change a research direction's title or description. Wording update	Title (up to 60 characters); description (up to 300)	Optional shorter description, on request	Only the four directions can be edited. Their ids cannot change	Approve	That direction's record changed
U11	Delete a project. Duplicate or cancelled	Choose the project	None	Shows the people whose profiles list it. Typed confirmation	Confirm	Record deleted; backup kept

B3. Events

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U12	Add an event from a notice. A notice arrives by email or on paper	Notice text (pasted or file)	Event fields as a structured object (title, kind, dates, venue) and a one- or two-sentence summary	Dates are real calendar dates. End on or after start. Kind from the list. Every date and venue in the notice must also appear in the notice (otherwise warning). Series from the approved list	Preview of the event card and page, with each drafted field highlighted	Event record created
U13	Change an event's date, time or place. Dates move	Choose the event; new values	Optional summary update	Date rules. Lists other pages that mention the old date or place	Each listed line, before and after	Changed lines saved; other lines listed for review
U14	Cancel or postpone an event	Choose the event; "cancelled" or a new date	Optional short notice for the summary	The template has no cancelled status. Until proposed change P17, the summary is changed	Confirm	Summary changed
U15	Past events leave the upcoming list. Happens by date	Nothing	None	The site decides at each build. The tool shows when the last build was made	None	Archive updated after the next build
U16	Delete an event	Choose the event	None	Typed confirmation	Confirm	Record deleted; backup kept

B4. News, announcements and insights

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U17	Add a news item, announcement or insight. Something is to be published	Kind; date; headline; two to four facts; optional full text	Summary up to 300 characters, and body of 100 to 250 words, from the facts	Headline up to 120 characters. Date valid. Summary present. Nothing beyond the facts	Preview on the list and on the item's page	Record created; homepage updated at the next build
U18	Edit or delete a news item. A correction or a withdrawal	Choose the item	Optional rewrite	As U17	Changed lines, or a typed confirmation for delete	Record changed or deleted

B5. Job openings

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U19	Add an opening. A position is approved	Title; type (seven choices); deadline; summary facts; apply email (blank means the centre's general address)	Summary up to 400 characters, from the facts	Type from the list. Deadline a real date. Email valid. Warns if the deadline has passed	Preview	Record created
U20	Close an opening. Post filled or deadline passed	Choose the opening; set status to closed	None	Warns if the deadline has passed and the status is still open	Confirm	Status set to closed; listed as "Applications closed"
U21	Reopen or extend an opening. Deadline moved	New deadline; status open	None	Deadline is today or later	Confirm	Record changed

B6. Publications

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U22	Add a publication from a DOI or a pasted reference. A paper is accepted or published	DOI, or the pasted reference	Fields taken from the pasted text only: authors in order; type from the list	DOI cleaned (web prefix and "doi:" removed). Format checked. Duplicate DOI blocked. Year from 1950 to next year. At least one author. Optional online lookup, off by default	Preview of the entry, as it will appear	Record created
U23	Add a publication without a DOI (preprint, thesis, working paper)	Title; authors; venue; year; type; web link	None	Link is https. Year and authors checked	Preview	Record created
U24	Fix a DOI that does not resolve. A check shows it is broken	New DOI	None	Format. Resolution if online	Confirm	DOI changed
U25	Remove sample publications before launch	Choose items	None	Shows the "[Sample]" title on each	Confirm	Records deleted

B7. Partners and funded programmes

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U26	Add a partner. A new collaboration starts	Name; country; academic or industry; present or former; summary; featured yes or no	Summary up to 200 characters, from the facts	Name required. Featured partners show on the homepage only when present. Warns above six featured	Preview	Record created
U27	Mark a partner as former. A collaboration ends	Choose the partner	None	Warns if the partner is named in a funded programme	Confirm	Status changed
U28	Add a funded programme. A grant is awarded	Title; funder; partner (pick or type); amount as the funder writes it; period; summary	Summary from the facts	Warns if the partner is not on the partners list. Amount and period kept as text	Preview	Record created
U29	End a funded programme. The programme has finished	Choose the programme	None	No status field exists (proposed P16). Until then the only route is delete	Confirm	Record deleted

B8. Facilities

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U30	Add or edit a facility. New or changed equipment	Name; kind (pick or type); summary; order	Summary up to 200 characters, from the facts	Name and summary required	Preview	Record created or changed
U31	Remove a facility. Equipment retired	Choose the facility	None	Typed confirmation	Confirm	Record deleted; backup kept

B9. Documents (PDF files)

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U32	Upload a PDF. A report or policy is approved	PDF file; title; kind; date; description	None	The file really is a PDF (checked from its start, not its name). Size limit (proposed 20 MB). File name cleaned. Title and description present. Warns that PDFs are not checked for accessibility	Preview of the download entry; file name	PDF saved in the documents folder; document record linked

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U33	Replace or remove a PDF. A new version, or withdrawal	New file, or remove	None	As U32. The old file kept in the backup	Confirm	File replaced or removed

B10. Gallery albums

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U34	Add or edit an album (text only). An event is planned	Title; period; description	Description up to 300 characters, from the facts	Title and description required	Preview	Album created or changed
U35	Add photographs to an album. Photographs are supplied	Photographs, captions, consent	None	Not supported by the template. The tool says so and writes nothing until proposed change P15 is made	Not available in version 1	Nothing written

B11. Site settings and contact details

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U36	Change contact details. The email, phone or address changes	Email; phone; two address lines	None	Email valid. Phone in an accepted format. Display and dialling versions produced from one entry. Warns that the privacy notice must be reviewed if the address changes	Confirm	Settings changed
U37	Change the institute name or short names	Names	None	Not blank, and not still “Institute name”. Warns if the name appears in legal text	Confirm	Settings changed
U38	Change the student club link	Web address; label	None	Address must be https. Warns on example.org or GitHub addresses	Confirm	Link changed
U39	Set the domain. Final address confirmed	Domain	None	https. Both places changed together	Technical contact confirms	Two values changed

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U40	Change a navigation or footer label	Label; target page (dropdown of existing pages)	None	Target must exist. At most eight navigation items. Labels up to 16 characters (proposed)	Confirm	Changed

B12. Page wording

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U41	Change homepage text. A refresh of the wording	Headline (up to 80 characters); lead (up to 300); headings; button labels	Optional lead rewrite, from the facts	Length. No placeholder text left. Claims listed for the approver	Preview of the homepage	Text changed (needs P1)
U42	Change About text: mission, history, milestones, leadership	Paragraphs; milestone year, title and text (text up to 200 characters); leadership name chosen from People	Optional mission rewrite, from the facts	Leadership names must match People entries (needs P4)	Preview	Text changed
U43	Edit the Join page questions and answers	Question; facts for the answer	Answer up to 80 words, from the facts	No eligibility, fee or deadline claim that is not in the facts	Approve	Answer changed; “Sample” label removed (needs P10)
U44	Edit Support text: giving routes, tax notes, donor tiers	Wording approved by the finance office	None. The model is not used for finance text	Only non-empty text. Warns if placeholder wording remains	Finance approval recorded (name and date)	Text changed
U45	Edit contact enquiry types and the campus note	Text	Optional rewrite	Length	Confirm	Changed
U46	Edit the legal pages (accessibility, privacy, terms). Institute review	Text; values for each bracketed field	None. The model does not draft legal text	Lists bracketed fields still open. The “Draft for review” notice cannot be edited or removed by the tool	Institute approval recorded (name and date)	Text changed. Notice removed only by the technical contact
U47	Change the workshop series title and lead	Title; lead	Optional rewrite	Length	Confirm	Changed

B13. Samples and placeholders

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U48	Review every sample entry before launch	Nothing	None	Lists every sample entry by area, and the pages that show the Sample label	For each: replace, or remove	Records updated

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U49	Remove all samples in one step	Confirm the count	None	Blocks if a remaining entry depends on a sample (for example a project that names a sample person)	Typed confirmation	Samples removed; backup kept
U50	Find remaining placeholder text	Nothing	None	Scans records and page wording for the markers listed in EDITABLE_SECTIONS, area 17	Fix each one	List cleared

B14. Check, preview and undo

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U51	Run the site check after changes	Nothing	None	Runs the configured check command. Turns each error into a plain sentence naming the file and field	Reads the result	Pass, or a plain error with file and field
U52	Preview the site	Nothing	None	Needs a successful build. If there is none, says so	None	Local preview opens in the browser
U53	Undo a change. A mistake is noticed	Choose the run	None	Refuses if the file has changed since the run	Confirm	Files restored; the undo is logged

B15. Launch preparation and handover

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U54	Prepare the launch. A launch date is set	Approval records for content, legal and finance	None	Runs the template’s launch check plus the extra checks in section (e). Lists blockers and warnings	Approval boxes ticked and recorded	Readiness report saved
U55	Hand over the built site. Launch is ready	Nothing	None	Creates an export folder with the build and checksums. Does not contact any server	Technical contact deploys the folder	Export folder created

B16. Writing helper (the model)

#	Use case and trigger	Operator input	Model drafts	Tool checks	Operator approves	Result
U56	Check the writing helper. Start of a session, or after an error	Nothing	None	Server reachable. Model installed. A short test reply and its time	None	Status shown on the start screen

(c) Edge cases and failure modes

Each row names something that can go wrong and what the tool must do. Behaviour words: **Block** (refuse to save), **Warn** (show, let the operator continue), **Ask** (the operator must choose), **Fallback** (a safe default), **Log** (record in the history).

#	Situation	What can go wrong	Required behaviour
E1	A date with no year or in an unclear form (for example “10/03/2026” or “next Tuesday”)	Read as the wrong day or month. Some software reads 10/03 as 3 October	Block until the operator picks a date from a calendar. The model returns ISO dates (YYYY-MM-DD) only. The tool never guesses
E2	The notice gives no date for an event or news item	A required field is empty	Block; Ask the operator for the date
E3	End date before start date	The template does not check this	Block
E4	A multi-day event, or a one-day event	Dropped from “upcoming” at midnight UTC on its last day, which is early morning in India, for example	Keep the event upcoming through its whole last day in the centre’s time zone (needs P8). Until then, the start screen lists events that end today or tomorrow
E5	The site is not rebuilt after dates pass	“Upcoming” is stale	Warn on the start screen when the last build is over one day old and an event has passed. Offer to run the build
E6	Two people with the same name	Project links and profiles go to the wrong person	Block duplicates (capitals and spacing ignored). Ask for a unique display name if two real people share one
E7	A student name typed differently from the person’s name	The link and the profile entry silently disappear	Warn with the closest matches. Ask the operator to choose
E8	A person has no portrait	Initials are shown	Allow. The preview shows the initials
E9	A portrait path is wrong or the file is missing	A broken picture, with no build error	Block. The tool writes the photo path only after the file is in place. The launch check lists missing files
E10	A portrait is too small or the wrong shape	Blurred or cropped faces	Block below 800 px tall. Warn outside about 4:5 and offer a crop
E11	A very large photograph (for example a phone photo of 10 MB or more)	Slow page; upload fails	Resize to 1,000 px tall and keep the original in the backup (proposed). Block an original above 20 MB (proposed)

#	Situation	What can go wrong	Required behaviour
E12	A phone photograph is stored sideways	Portrait shown sideways	Apply the rotation stored in the file when importing (proposed)
E13	Alternative text for images	Visitors using screen readers hear nothing or a file name	Portraits use the person's name (template rule). Gallery photographs, once supported, are blocked until a caption and alternative text are written
E14	A very long title or name	Heading spreads over many lines; the browser tab title is cut at about 68 characters	Warn at 70 characters. Block at 120
E15	A very long unbroken text (a web address or DOI)	Overflows on a phone	Allow (the template wraps it). The tool shows a phone-width preview (proposed)
E16	A navigation label that is too long	The header overflows on a desktop screen	Block over 16 characters (proposed)
E17	The model writes a fact that was not given (a name, number, date or link)	False information looks plausible and gets published	Fact check: every number, date, name and link in a draft must appear in the operator's input. Otherwise the tool highlights it and blocks until it is removed or confirmed
E18	The model's reply is not in the right format	Bad data reaches the save step	Retry once with stricter instructions. Then show an empty form with what could be read. Nothing is saved from a failed reply
E19	The model is slow or runs out of memory	The computer freezes or the tool crashes	Time limit (default 300 seconds). A cancel button. Warn not to run the build and the model together on an 8 GB computer
E20	The model server is not running	Drafting fails	Plain message; manual entry still works. Nothing crashes
E21	The model server address points to another computer	Notices and CVs leave the computer	Block unless the technical contact has set the address. Show the address on the Settings screen
E22	The operator asks for legal, tax, finance or medical wording	Legal risk	Not offered. The tool says the institute must write this text
E23	A DOI pasted with a web prefix or "doi:" in front	Broken link	Clean the DOI and show the cleaned value before saving
E24	A DOI that does not resolve, or uses the placeholder prefix 10.0000	Broken link on the publications page	Warn "DOI not confirmed". Allow saving with a reason noted. Never block for lack of internet
E25	The same DOI is already listed	A duplicate entry	Block. Show the existing entry
E26	A publication has both a DOI and a web link	Only the DOI is shown; the link is hidden	Warn that only the DOI will show. Ask which to keep
E27	The check fails after a save	The site cannot be built	Keep the saved change. Show the error in plain English. Offer Undo. Mark the site "not ready" until it passes

#	Situation	What can go wrong	Required behaviour
E28	The build error names a file or field the operator does not understand	Confusion and guesswork	Translate common messages (missing field, wrong choice, invalid date, invalid address) into plain instructions. Keep the original message in a “details” box for the technical contact
E29	The HTML check fails (markup problem)	A page may be broken	Show the page and the rule. No automatic fix. The technical contact decides
E30	Node is missing or older than 22.12	The build cannot run	Block the build with a plain message. Editing still works
E31	Dependencies installed on another operating system (for example a folder copied from Windows to WSL2)	The build fails in confusing ways	Detect the mismatch (a marker file records the system). Ask the technical contact for a fresh install
E32	Two people edit the same file at the same time	One overwrites the other	Compare the file’s fingerprint at load and at save. If it changed, Block and offer to reload
E33	A file is open in another program (Windows)	A save fails midway	Retry three times over a few seconds. Then a plain message (“close the file in other programs”). Never leave a half-written file
E34	The site folder is inside a cloud-synced folder (OneDrive, Dropbox, Google Drive)	Locked files; duplicate copies created by the sync	Warn at start-up. Recommend a local folder (section f)
E35	Power cut or crash during a save	A corrupted file	Write to a temporary file, then replace the original in one step
E36	A name that Windows reserves (CON, PRN, AUX, NUL, COM1 to COM9, LPT1 to LPT9)	The file cannot be created	Block. Add a word to the name
E37	Two names that differ only in capitals (Jane-Doe and jane-doe)	Collide on Windows	The uniqueness check ignores capitals
E38	Accented letters or dashes in names and titles (Müller, “human–robot”)	File names break or look wrong	Keep the text as typed in the file, saved as UTF-8. Make the file name plain ASCII (for example “muller”)
E39	A long folder path on Windows	Tools fail when the path passes 260 characters	Check the path length at start-up. Recommend a short folder near the drive root
E40	An entry is renamed after it is published	Its web address changes and old links break	Block renaming. Offer to create a new entry instead. A redirect is needed for a true rename (proposed P19)
E41	An entry that other pages depend on is deleted (a student, a partner named by a funded programme)	Pages lose links or show blanks	Warn and list the dependants. Delete only after confirmation
E42	“[Sample]” is left in a title after the sample is replaced	Visitors see “Sample”	The tool strips the prefix when a sample is replaced. The launch check lists any left

#	Situation	What can go wrong	Required behaviour
E43	Bank account details or private personal data typed into a text field	The data is published	Block strings that look like bank account numbers or IBANs. Warn on personal phone numbers and email addresses
E44	A photograph without consent	A privacy breach	Block until the consent box is ticked. The consent is recorded with the change
E45	An opening's deadline has passed but its status is still open	The position still shows "Apply by email"	Warn, and offer to close it. The template will not close it by itself
E46	More than four featured projects, or more than six featured partners	Extra items are hidden without notice	Warn. Show which items will appear
E47	The address or phone number changes	The privacy notice and the footer disagree	Warn that the privacy notice must be reviewed. Both phone fields always change together
E48	Homepage counts include sample entries	Wrong numbers are shown	Warn on the start screen. Proposed P18 counts only published entries
E49	Run logs keep notices and CVs	Personal data kept on the computer	Keep logs for 90 days (proposed) and provide a purge. Logs are never sent anywhere
E50	A settings file has an unknown key or a typo (for example "modle")	The wrong model runs without warning	Block at start-up and name the key (existing rule D10)
E51	The template is upgraded without updating the tool	The tool follows old rules	Compare a fingerprint of the template's rules at start-up. Block saves until the tool is updated
E52	A new model or new prompt is introduced	Quality drops without anyone noticing	Run a fixed test set (sample notices with known answers) before a new model is used
E53	The backup folder fills the disk	Saves fail	Keep the last 200 runs (proposed). Warn when free space is low
E54	An undo after the file was changed by someone else	Newer work is overwritten	Refuse the undo if the file changed since the run (existing rule S4)
E55	Two copies of the tool run on the same site	Two write paths	A lock file in the site's state folder. The second copy refuses with a plain message
E56	A file renamed to .pdf that is not a PDF	Broken download	Check the file's first bytes. Block if it is not a PDF

(d) User interface requirements (Streamlit, version 1)

Principles

- One screen for one task. The operator sees exactly what will change before anything is saved.
- Every fixed choice is a dropdown. Typing is for facts the operator knows.
- Every save goes through review and then approval. No silent writes.
- Plain English on every operator screen. Technical details appear only in a "For the technical contact" box.

- The undo button is visible on every result screen.
- The screens call only the service facade (section e). They contain no site rules.

Screens and form fields

#	Screen	Fields and controls, in order	Limits and help text
1	Start	Panels (read-only): writing helper status with a “Check” button; site folder; last check result and its age; count of sample entries; list of the three most important warnings	Shows “last build was N days ago” when old
2	People	Table with a group filter; buttons Add, Edit, Move to alumni, Delete. Form: Name (80); Group (dropdown, six); Role (80); Department (120); Interests (tags, up to 10, 60 each); Homepage (https web address); Portrait (upload, JPG or PNG, 20 MB; resized on import); Consent (tick box, required with a portrait); Order (number, 1 to 999); Facts for the profile (text, 1,000 characters); Biography (generated at 80 to 150 words; editable up to 250)	Help: “Write the name as it should appear on the website”
3	Projects	Table grouped by direction. Form: Title (120); Direction (dropdown, five); Students (multi-select from People, plus “type a name” with a warning); Status (optional dropdown); Summary (300); Featured (tick box, warns above four); Order; Abstract (generated, 120 to 250 words)	Warning shown when a student is not in People
4	Research directions	Four cards. Form: Code (4); Title (60); Description (300)	Help: “The four directions are fixed by the website’s structure”
5	Events	Tabs: Upcoming; Past; Add from a notice. Notice tab: Notice text (3,000 characters); “Read notice” button. Event form: Title (120); Kind (dropdown, three); Series (dropdown: none or the approved list); Start date (calendar); End date (calendar, not before start); Venue (120); Summary (300); Description (generated, 60 to 200 words)	Warnings for any date or place not found in the notice
6	News	Table. Form: Kind (dropdown, three); Date (calendar); Headline (120); Summary (300); Full text (generated, 100 to 250 words)	

#	Screen	Fields and controls, in order	Limits and help text
7	Job openings	Table grouped by open and closed. Buttons: Close now; Reopen. Form: Title (120); Type (dropdown, seven); Status (open or closed); Deadline (calendar, optional); Summary (400); Apply email (optional); Order	Warning when the deadline has passed
8	Publications	Add-from-DOI panel: DOI; “Look up” button; optional reference text. Form: Title (250); Authors (one per line, in order, at least one); Venue (120); Year (number); Type (dropdown, six); DOI; Web link (optional); Direction (optional)	DOI check result shown in plain words
9	Partners	Table in three groups: academic, industry, former. Form: Name (120); Country (60); Kind (dropdown, two); Status (dropdown, two); Summary (200); Featured (tick box); Order	
10	Funded programmes	Table. Form: Title (120); Funder (120); Partner (pick from list or type); Amount (60, “as the funder writes it”); Period (30, for example 2024–2027); Summary (300); Order	
11	Facilities	Table. Form: Name (120); Kind (pick or type); Summary (200); Order	
12	Documents	Table. Form: Title (120); Kind (dropdown, five); Date (calendar, optional); Description (300); PDF (upload, PDF only, 20 MB)	Shows whether a PDF is attached
13	Gallery	Table of albums. Form: Title (120); Period (30); Description (300); Order	Notice: “Photographs are not supported yet”
14	Page wording	One section per page in EDITABLE_SECTIONS, area 14. Each section: its fields (text boxes) and a “Preview page” button	Length limits per field (proposed)
15	Contact and names	Name; short name; institute; institute short name; institute website (https); email; phone; address (two lines); student club label and link	Domain is not on this screen (technical)
16	Navigation and footer	Navigation items: label (16) and target (dropdown of existing pages). Footer: link labels and targets within the fixed columns	At most eight navigation items
17	Legal pages	For each of the three pages: its text sections; each bracketed field as its own box; approval record (approver name and date)	The “Draft for review” notice is shown but cannot be edited on this screen

#	Screen	Fields and controls, in order	Limits and help text
18	Samples and placeholders	List of sample entries by area; placeholder scan results; buttons Replace, Remove, and Remove all (with count)	
19	Check and preview	“Check the site” button; results in plain English with a details box; “Open preview” button; “Export for technical contact” button	Shows the last result and its time
20	History	List of runs (date, what changed, who approved); “Undo last change” button; details	
21	Launch	Launch checklist in plain English; status per item; approval boxes; “Save readiness report” button	Does not publish anything
22	Settings (technical)	Site folder; writing helper address and model name; time zone for events; check command (fixed, shown only); upload limits; log retention	Locked by a technical password (proposed)
23	Help	How to use each screen; a short glossary; who to contact	

Behaviour shared by all screens

- Forms are submitted as a whole (Streamlit forms), so half-finished entries are not saved.
- Long tasks (drafting, the build) run in the background with a progress message and a cancel button.
- Uploads are checked on the server, not only in the browser. Size limits are set in the Streamlit configuration for each type.
- The tool listens only on the local address (127.0.0.1) and says which address to open.
- Drafts are kept in the state folder so they survive a restart.
- Text uses British English. Error messages say what to do next.
- The tool’s own screens meet WCAG 2.2 AA contrast and can be used from the keyboard (proposed).
- Previews are shown at phone width as well as desktop width (proposed).

(e) Service architecture

Layers

```
Screens (Streamlit)            replaceable; any other front end can replace this layer
      |  calls only
Service facade (api)           one function per operator action; returns plain results
      |
Services                       people, projects, research, events, news, openings,
                               publications, partners, funding, facilities, documents,
                               gallery, site settings, page wording, legal pages,
                               samples, placeholders, launch, drafting
      |  use
Rules                          schema rules, references, dates, DOI and web addresses, slugs
      |  use adapters
Adapters                       content store, model client, build runner, media,
                               history, configuration, platform (Windows and WSL2)
```

Modules (one responsibility each)

Module (proposed name)	One responsibility	Main interface	How it stays replaceable
`ui`	Show screens and collect input	Streamlit pages only	Imports only `api`. Contains no rules
`api`	One function per operator action	For example `preview_person(form)`, `approve(preview_id)`, `run_check()`. Returns plain results	The single entry point. Another front end calls the same functions
`config`	Read settings; reject unknown keys	`load_settings(folder)`	Exists today (`config.py`)
`content_store`	Read and write content files safely	`list(collection)`, `read(collection, id)`, `write(collection, id, record, body, expected_fingerprint)`, `delete(...)`, `fingerprint(...)`	An interface (`ContentStore`). A database could replace the files later
`frontmatter`	Change only the changed lines of a file's top block	`set_values(text, changes)`, `render(record, body)`	Exists today (`frontmatter.py`)
`schema_rules`	The template's rules per collection, with proposed limits; a fingerprint of the template's rules	`validate(collection, record)` returns plain messages	A table of rules, checked against the template at a pinned commit
`references`	Checks between entries and files	`check_students(names)`, `check_partner(name)`, `check_files()`, `dependants(entry)`	Works through `content_store`
`slugs`	Short names for file names and web addresses	`make_slug(title, taken)`, `is_reserved(name)`	Pure functions, tested with Windows reserved names
`dates`	Strict date reading; the upcoming rule	`parse_iso(text)`, `is_upcoming(event, now, zone)`	The time zone comes from settings

Module (proposed name)	One responsibility	Main interface	How it stays replaceable
`identifiers`	DOI and web-address rules	`clean_doi(text)`, `check_doi(doi, online)`, `check_url(url)`	The online lookup is an optional adapter, off by default
`media`	Portraits and PDFs: checks, resizing, copying	`import_portrait(file, slug, consent)`, `import_pdf(file, slug)`	Image handling behind an interface
`drafting`	Build the prompt for each task, call the model, check the result	`draft(task, facts)` returns a draft with warnings. Never writes	Prompts are text files. The model is reached through `model_client`
`model_client`	Talk to the local model server	`health()`, `complete(messages, json_mode)`	Adapters: Ollama (exists) and an OpenAI-compatible server (exists)
`page_wording`	Read and write fixed page wording	`sections()`, `get(key)`, `set(key, text)`	Works on a data file after P1. Until then, a fixed map of keys to lines, tested
`site_settings`	Read and write site settings	`get()`, `set(...)`	A data file after P1. Until then, a strict editor for the TypeScript file
`legal_pages`	Legal text, bracketed fields and the protected notice	`open_fields(page)`, `set_field(page, field, text)`	The notice text is protected by a test
`samples`	Find, replace or remove sample entries	`list()`, `replace(...)`, `remove(...)`	Uses `references` to find dependants
`placeholders`	Scan for placeholder markers	`scan(site)`	Marker list kept in one file
`build`	Run the check command and read the result	`run_check(timeout)` returns messages mapped to file and field	The command comes from settings only (existing rule S8)
`launch`	Build the launch report	`report()` returns blockers, warnings and passes	Reads the template's launch check output and adds its own checks
`history`	Runs, backups, undo, approvals	`list_runs()`, `restore(run_id)`, `record_approval(...)`	Exists in part (`apply.py`); split out
`platform`	Paths, line endings, file locks, long paths, Node and model detection	`to_site_path(...)`, `write_text(path, text)`, `check_environment()`	One adapter per operating system
`export`	The handover folder with checksums	`export(folder)`	

Rules for replaceability

- The screens and `api` never import an adapter directly. The adapters are passed in at start-up in one place.
- A content type is data: a rule table, a folder, and a template for showing it. Adding one should not need a new class.
- Storage, the model and the screens can each be replaced without changing the rules.
- The rules are tested against example records with no model and no network.
- Every message the operator sees comes from one file, so wording can be changed without code changes.

Existing code to keep and extend

- Keep: `config.py`, `frontmatter.py`, `validate.py`, `apply.py`, `workflows.py`, `events.py`, `llm/` (Ollama and OpenAI-compatible clients), `prompts/`, `text_utils.py`, `plan.py`, `errors.py`.

- `cli.py` becomes one front end among others. The Streamlit screens are a second.
- `schemas.py` covers only events today. It must cover every collection (through the rule table).

Changes to existing rules (proposed)

- **Write permissions.** The default writable paths (`site-agent.example.toml`: `src/**/*.md`, `src/**/*.astro`, `src/**/*.ts`, `src/**/*.tsx`) must be narrowed to content and media only: `src/content/**/*.md`, `public/people/**`, `public/documents/**`, and the settings data file once P1 creates it. Code files must never be writable by the operator tool. It would supersede decision D7 in `docs/DECISIONS.md`, once agreed.
- **Atomic writes.** Writes are direct today (`site.py` line 68 writes straight to the file). Use a temporary file and a single replace, with retries on Windows.
- **Shell command.** The check command runs through the shell (`apply.py` lines 116-118, `shell=True`). On Windows that is cmd.exe; on WSL2 it is sh. Keep one configured command and test it on both.
- **Template state folder.** The tool asks that `.site-agent/` be listed in the website's `.gitignore`. The template's `.gitignore` does not list it (add it, P12).
- **Missing documents.** The tool's README links to `docs/PLANNING.md`, `docs/MODELS.md`, `docs/TEMPLATE_CONTRACT.md` and `CONTRIBUTING.md`. None of them exists in the repository. `docs/ARCHITECTURE.md` links to PLANNING.md. These must be written or the links removed.

Template changes needed (proposed)

These keep every page's layout and structure. They move wording into data files, make placeholders visible to the launch check, and remove silent breakages.

ID	Change	Why	Priority
P1	Move page wording (EDITABLE_SECTIONS area 14) and site settings (area 13) into data files that pages read	Most wording lives in code files	Must
P2	Give research directions, partners and funded programmes a sample flag, and add a placeholder scan to the launch check	Placeholders there are invisible today	Must
P3	Replace the literal series name `series-1` (`events.astro` 21; `index.astro` 31) with a fixed list of series	Any other series name is silently dropped	Must
P4	About page: read facilities and leadership from their collections, or remove the duplicates	Two copies of the same facts	Should
P5	Use the site email on the people page (line 140) and the gallery page (line 45)	Hard-coded addresses	Must
P6	Make the header monogram read the short name; set the domain in one place; make the displayed and dialling phone numbers come from one entry	Two sources for one value	Should
P7	Display the social-links setting, or remove it	A setting that does nothing	Should
P8	Events: block end dates before start; keep an event upcoming through its last day in the centre's time zone	Wrong cut-off	Must

ID	Change	Why	Priority
P9	Build-time checks: DOI without a web prefix; photo and document paths must exist (missing file fails the build); https-only club and homepage links	Broken links pass silently	Must
P10	Show the Sample label on every people list and on homepage faculty cards (`index.astro` 143). Show the FAQ label only on sample answers	Mislabelled content	Should
P11	Publications: show the web link even when a DOI exists, or document the rule	A link is hidden	Should
P12	Add `.site-agent/` to the template's `.gitignore`	Tool files would enter version control	Must
P13	Launch check: fix the message path (`docs/06` should be `docs/03`); check the domain in both files; check the institute name, email, phone and address for placeholder values	Launch check misses real problems	Must
P14	Decide the licence and add a licence file (the template has none)	Legal position unclear	Must before sharing
P15	Gallery: a photo list with captions and alternative text. Partner logos as an option	Photographs cannot be added	Should, before the first event photographs
P16	Funded programmes: a status (current or finished)	A finished programme cannot be hidden	Should
P17	Events: a status (cancelled or postponed) and a notice line	A cancellation cannot be shown	Should
P18	Homepage and page counts: count only published, non-sample entries	Counts are wrong while samples exist	Should
P19	A redirect list, so renamed or removed pages keep working	Old links break	Should

P1, P2 and P3 should be agreed before the operator tool is built. Otherwise the tool can only edit what is already in data files (people, projects, events, news and similar), and the placeholders stay invisible.

(f) Cross-platform requirements (native Windows and WSL2)

Topic	Requirement	Applies to	Evidence
Operating systems	The tool and the site build must both run on native Windows and on WSL2 (Ubuntu). Test each on the actual machine before the bundle is committed	Both	The tool's README says Windows has not been tested. The template says to use WSL (`README.md` 33; `docs/03` 13)

Topic	Requirement	Applies to	Evidence
Node	Node 22.12 or newer, checked at start-up. On Windows, use a portable (zip) Node 22 build so no installer or administrator rights are needed (technical contact to confirm)	Both	`package.json` lines 7-9
Platform-specific packages	Do not copy `node_modules` or a Python environment between Windows and WSL2. Install on each machine. Each operating system has its own compiled files	Both	`package-lock.json` lists compiled packages for win32, linux and darwin (for example `@astrojs/compiler-binding-win32-x64-msvc` and `…-linux-x64-gnu`)
Paths	Store paths relative to the bundle. Never write `C:\` or `/mnt/c` into settings. Convert paths in one platform adapter	Both	`astro.config.mjs` lines 14-16
Where the site lives	On WSL2, keep the site on the Linux file system, not on a Windows drive (`/mnt/c`, `/mnt/d`). File-change events are unreliable there, which is why the template polls	WSL2	`astro.config.mjs` lines 14-16
Line endings	Write LF line endings and UTF-8 without a byte-order mark. Read files with or without the mark	Both	`.gitattributes` line 2 (`text=auto eol=lf`). `site.py` line 68 writes LF; line 49 reads `utf-8-sig`
Long paths	Windows limits paths to 260 characters by default. Keep the site folder near the drive root (for example `C:\CIR\site`). Check the length at start-up. Enabling long paths needs administrator rights (technical contact)	Windows	General Windows behaviour; a deep `node_modules` tree is likely to exceed the limit. To verify on the target machine
File names	Apply the reserved-name, trailing-dot and capital-letter rules (section c, E36 and E37)	Windows	`slugs` module
File locks	A file open in another program, or held by a preview server, can block saves and replacements. Retry, then give a plain message. Build into a new folder and then swap (proposed). Do not write into `dist/` while the preview runs	Windows	No retry exists today (`apply.py`). The Astro dev server watches files (`astro.config.mjs` 14-16)
Synced folders	Do not keep the site in OneDrive, Dropbox or Google Drive folders. Warn at start-up	Windows	General behaviour (section c, E34)
Memory	On an 8 GB computer, do not run the build and the model at the same time	Both	General. To verify by measurement

Topic	Requirement	Applies to	Evidence
Model server	Default address `http://localhost:11434`, set in settings. The tool checks it is reachable and the model is installed. On WSL2 with Ollama on Windows, test the address that the bundle uses. Never start an unknown program automatically	Both	`llm/ollama.py` line 26 (`/api/chat`); `site-agent.example.toml` `base_url`
Model files	Models are large and Ollama keeps them in its own folder, whose location differs by system. The bundle holds a manifest (model name and checksum), not the model	Both	General Ollama behaviour
Local address and firewall	Streamlit listens on 127.0.0.1 only. The Windows firewall may ask for permission; allow private networks only	Windows	Section (d)
Administrator rights	The tool runs as a normal user. Installing Node, Ollama or long-path support may need an administrator (technical contact)	Windows	General
Check command	The check command must work the same in cmd.exe and sh. Keep it simple and test on both	Both	`apply.py` 116-118
Text encoding	Names with accents and dashes are UTF-8 in files and in the console	Both	`cli.py` line 222 sets UTF-8 on the output stream
Launchers	Use `.bat` files or a Python launcher. Avoid PowerShell scripts (execution policy) and avoid shell-only launchers	Both	General
Antivirus	Unsigned programs may be blocked by Windows SmartScreen. Provide signed files, or a documented way to allow them	Windows	General

Bundle requirements

- **What the bundle holds:** the tool (a pinned version); a Python environment created on each machine; the site folder (the template at its pinned commit, with the approved changes); a settings file with relative paths only; prompt files; the operator guide; a model manifest; optionally a portable Node 22 for each system.
- **What the bundle does not hold:** `node_modules`, Python environments, and the model files. These are created or pulled on each machine.
- **Two builds, one source:** one bundle for Windows and one for WSL2, built from the same source with the same version numbers.
- **Moving between machines:** copy the content, backups, run logs and settings. Do not copy `node_modules`, Python environments or Node. The technical contact runs the set-up step on the new machine.
- **Start-up checks:** Node version; Ollama reachable; model installed; site folder found; the site's dependencies match the system; path length; cloud-sync warning; template fingerprint.
- **Upgrades:** replace the tool folder; keep the state folder (backups, runs, drafts) and the content.
- **Pinning:** the Python dependencies are pinned in a lock file (proposed) so a rebuilt bundle behaves the same.

(g) Model requirements

Tasks, outputs and context

Context means the text the model reads plus the text it writes. The sizes below are targets to be measured on the actual computer, not measured results.

Task	Input	Output format	Output limit	Context needed (target)	Checks after the model
T1 Notice to event fields	A notice, usually 50 to 300 words	One structured object with fixed keys	About 300 tokens	2,048 (3B) to 4,096 (8B)	Format; every date, place and name appears in the notice; kind from the list
T2 Event summary and description	Event fields and facts	Summary: one or two plain sentences. Description: 60 to 200 words, plain paragraphs	About 80 tokens (summary); about 300 (description)	3,072	Length; no new facts
T3 Person biography	Three to six facts	Two to four plain paragraphs. No headings, no links	About 300 tokens	3,072	80 to 150 words; no name, year, number or link not in the facts
T4 Project summary	Title, students, direction, facts	One or two sentences, up to 30 words	About 60 tokens	2,048	Length; names match People
T5 Project abstract	Title and notes, up to 1,000 words	120 to 250 words, plain paragraphs	About 500 tokens	4,096 (3B: keep notes short); 6,144 (8B)	Word count; facts check
T6 News item	Facts or notes	Headline up to 120 characters; summary up to 300; body 100 to 250 words	About 600 tokens	4,096	Lengths; facts check
T7 Opening summary	Notice text, up to 600 words	Plain text up to 400 characters	About 150 tokens	3,072	Length. Deadline and contact come from fields, not the model
T8 Publication from a pasted reference	Reference text, up to 300 words	Structured object: title, ordered authors, venue, year, type from the list, DOI	About 400 tokens	3,072	DOI and year checked by code; authors compared with the pasted text; type from the list
T9 Profile from a CV	CV text, one or two pages	Structured object: name, role, department, interests	About 300 tokens	6,144 (chunk longer CVs)	Operator confirms each field. The name must appear in the CV
T10 Shorten to a limit	Existing text and a limit	Plain text	About one third of the limit	2,048	Length; no new facts (word overlap check)
T11 FAQ answer	Question and facts	Plain text up to 80 words	About 150 tokens	3,072	No eligibility, fee or date claim not in the facts

Not model tasks. These are code or human tasks: DOI and web-address format; dates; file and web names; numbers; phone and email format; legal, finance, donor and tax text; bank details; photo consent; image handling; alternative text for

photographs; approval records.

Settings. Temperature 0.1 for structured extraction (current default in `site-agent.example.toml`). About 0.3 for drafting (proposed). Context window 8,192 as the default ceiling; the setting costs memory, so measure it. Time limit 300 seconds (current default).

Choosing a model by memory

Computer memory	Realistic choice	Notes
8 GB	A 3B model, quantised	Expect slow replies. Do not run the build and the model together. Manual forms must always be available
16 GB	A 7B or 8B model. The current default is `qwen2.5:7b`	Better extraction from longer notices and CVs. Still needs every check

Set a target for response time on the target machine (proposed): under 60 seconds for a biography on 16 GB, and under 120 seconds on 8 GB.

What a 3B model can do, and what it cannot

Can do, with the checks in place: - Turn three to six short facts into two or three plain sentences. - Shorten a sentence to a limit. - Pick one value from a short list. - Write a simple structured object from a short notice, when the server's JSON mode is on. JSON mode guarantees valid syntax, not the right fields, so the tool still checks the fields.

Cannot do reliably: - Follow more than a few rules at once. - Count words, or stay within a length without checking. - Avoid filling gaps with plausible invented details when the facts are thin. - Read a long CV or a multi-page notice in one go. - Read PDFs or images. - Understand unusual date formats. - Keep names and numbers consistent across a long text. - Judge accuracy, tone or legal risk.

What an 8B model adds: more reliable extraction from longer text, more consistent style, and fewer format failures. It still cannot be trusted without the checks.

Rules for every model

- The operator supplies the facts. The model is never the source of a fact.
- The tool checks every output (format, length, facts) before showing it. Model output is never saved directly.
- A failed reply is retried once with stricter instructions. After that the operator gets an empty form and the notes.
- The model never writes DOIs, web addresses, email addresses, phone numbers, dates to be calculated, bank details, or legal, finance, donor or tax text.
- Prompts are text files that the technical contact can improve without code changes (existing design).

(h) Open questions for the CIR team

Each question has a recommended answer. The team should confirm or change each one before the tool is built.

#	Question	Recommended answer
1	Which computer will the operator use: native Windows, or WSL2?	Test native Windows first, on the operator's machine, by installing Node 22 and running the check. Use WSL2 only if that test fails
2	Where will the site folder live?	On a local disk, with a short path (Windows: near the drive root; WSL2: on the Linux file system). Never in a synced or network folder
3	Which model, on which computer?	16 GB: the current default, `qwen2.5:7b`. 8 GB: a 3B model only after testing on real notices. Manual forms always available

#	Question	Recommended answer
4	Who approves wording and photographs?	A named content approver, for example the head of centre. The tool records the approver’s name and date with each batch
5	Who approves the legal pages?	The institute’s legal office, in writing. The tool records the approval. The technical contact removes the “Draft for review” notice
6	Who writes finance, tax and donor text?	The finance office. The model is not used for it
7	Will bank account details be published?	No, as the template says. Keep “account details on request”
8	What is the final domain, and who hosts the site?	Institute IT confirms both before the first real content. The technical contact sets the domain in both files. The launch check must check both (P13)
9	What happens to the student club link?	Point it to the club’s live site once it is up. Until then, hide it or mark it as pending, and keep the launch warning
10	What event series names are used?	A fixed list agreed by the centre, held by the tool, once P3 is done. Not the placeholder `series-1`
11	Which time zone decides “upcoming” events?	The centre’s local time zone. An event stays upcoming through its last day (P8)
12	Delete or archive?	No permanent delete in the tool. “Remove” moves the file to an archive folder outside the site, kept for 90 days (proposed). The technical contact deletes permanently
13	What consent is needed for photographs?	Written consent from each identifiable person. The tool records a yes or no with each change. The centre keeps the signed form
14	When will gallery photographs be supported?	Before the first event with photographs: approve P15 (photo list with captions and alternative text). Until then, albums are text only
15	Are partner logos wanted?	Text tiles until a partner gives permission for a logo. Then P15
16	Which languages?	English, with British spelling, for version 1
17	Where do notices, CVs and drafts go?	They stay on the computer. Local model only. Logs are deleted after 90 days (proposed)
18	Who decides the licence for the template and the tool?	The centre, before sharing. The template has no licence file. The tool’s MIT licence is only proposed
19	Who is the technical contact after handover?	A named person and a deputy, recorded on the centre’s handover sheet
20	Should homepage counts include samples?	No. Count published entries only (P18). Until then, hide the counts
21	Are the homepage limits right (four faculty, four featured projects)?	Keep them. The tool warns when more are chosen
22	Can projects name students who are not in People?	Yes, with a warning. The link appears only when the name matches a person

#	Question	Recommended answer
23	Should About leadership come from People entries?	Yes (P4), so each fact has one source
24	Is a contact or newsletter form wanted?	Not in version 1. Keep email. A form needs a decision on storing data and a privacy notice change
25	Who signs off the launch?	The content approver and the technical contact, recorded in the launch report